

Pre-screening

Participant	021
	Yes No
Do you or have you had a history of migraine headaches?	☐ ☒
Do you or have you had a history of claustrophobia?	☒ ☐
Do you or have you had a history of frequent or severe motion sickness?	☐ ☒
Do you or have you had a history of any health problems (e.g., seizures, diabetes, heart problems, vertigo) that affect your ability to drive?	☐ ☒
If you are a female, are you or is there a possibility that you might be pregnant?	☐ ☒

Demographics

1. What is your race/ethnicity?	☒ White / Caucasian
2. What is your gender?	☒ Female
3. How old are you?	32
4. What is the highest level of education you have completed?	☒ 12th grade or less
5. Which option best describes your current home?	☒ It is owned or being bought by you or someone in household.
6. How do you pay for healthcare?	☒ Private insurance
8. Which best corresponds to your current work situation?	☒ Not working or looking for work.
9. What is your combined family income for the past 12 months (from all sources)	☒ Do not know.

Timeline Followback

Visit 1

Visit 1 Date

07-23-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 2

Visit 2 Date

08-14-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 3

Visit 3 Date

08-28-2025

Have you used opioids or any other substances during the assessment period?

☒ Yes

Visit 4

Visit 4 Date

09-15-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Visit 5

Visit 5 Date

09-16-2025

Have you used opioids or any other substances during the assessment period?

☒ No

Phone

Participant # _____

021

System Usability Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently	☐	☐	☐	☒	☐
2. I found the system unnecessarily complex	☒	☐	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☐	☒	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☒
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☒	☐
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this app experience.	☒	☐	☐	☐	☐
2. The time I spent in this app just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the app task.	☐	☐	☐	☐	☒
4. I felt frustrated while using the app.	☒	☐	☐	☐	☐

5. I found this app to be confusing.	☒	☐	☐	☐	☐
6. Using this app was taxing.	☒	☐	☐	☐	☐
7. The app is attractive.	☐	☐	☐	☐	☒
8. The app is aesthetically appealing.	☐	☐	☐	☐	☒
9. The app appealed to my senses.	☐	☐	☐	☒	☐
10. Using this app was worthwhile.	☐	☐	☐	☒	☐
11. The therapy experience was rewarding.	☐	☐	☐	☐	☒
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Scale P

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were videos easy to understand.	☐	☐	☐	☐	☒
4. The breathing exercises were smooth and easy to do.	☐	☐	☒	☐	☐

AR

Participant ID _____

021

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Private insurance |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGROUP Presence Questionnaire

- | | |
|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Somewhat Disagree |
| 2. I felt that the virtual world surround me. | ☒ Completely Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Agree |
| 4. I did not feel present in the virtual space. | ☒ Neither Agree nor Disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something form the outside. | ☒ Somewhat Agree |
| 6. I felt present in the virtual space | ☒ Somewhat Agree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely Disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Somewhat Disagree |
| 11 How real did the virtual world seem to you? | ☒ Very unreal |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Completely Agree |
| 13. How real did the virtual world seem to you? | ☒ Very unreal |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat Disagree |

System Usability Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☒	☐	☐
2. I found the system unnecessarily complex.	☐	☒	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☒	☐
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☒	☐	☐
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☒	☐
8. I found the system very cumbersome to use.	☐	☒	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this augmented reality experience	☐	☒	☐	☐	☐
2. The time I spent in augmented reality just slipped away.	☒	☐	☐	☐	☐
3. I was absorbed in the therapy task.	☐	☐	☒	☐	☐
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☒	☐

8. The augmented reality is aesthetically appealing.	☐	☐	☐	☒	☐
9. The augmented environment appealed to my senses.	☐	☐	☐	☐	☒
10. Using this system was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☐	☒
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Evaluation AR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☐	☒
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
3. The videos were clear and easy to understand.	☐	☐	☒	☐	☐

VR

Participant ID _____

021

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Private insurance |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGroup Presence Scale

- | | |
|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Completely Agree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Agree |
| 4. I did not feel present in the virtual space. | ☒ Completely Disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Neither Agree nor Disagree |
| 6. I felt present in the virtual space. | ☒ Completely Agree |
| 7. How aware were you of the real world surrounding you while navigating in the virtual world? | ☒ Neither aware nor unaware |
| 8. I was not aware of my real environment. | ☒ Neither Agree nor Disagree |
| 9. I still paid attention to my real environment. | ☒ Neither Agree nor Disagree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Agree |
| 11. How real did the virtual world seem to you? | ☒ Very real |
| 12. How much did your experience in the virtual environment seem consistent with your real-world experience? | ☒ Neither Agree nor Disagree |
| 13. How real did the virtual world seem to you? | ☒ Very real |
| 14. The virtual world seemed more realistic than the real world. | ☒ Neither Agree nor Disagree |

System Usability Scale VR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☒	☐	☐	☐	☐
2. I found the system unnecessarily complex.	☐	☐	☐	☐	☒
3. I thought the system was easy to use.	☒	☐	☐	☐	☐
4. I think that I would need the support of a technical person to be able to use this system.	☐	☒	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☒	☐	☐
6. I thought there was too much inconsistency in this system.	☐	☐	☐	☐	☒
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☐	☐	☒	☐	☐
9. I felt very confident using the system.	☒	☐	☐	☐	☐
10. I needed to learn a lot of things before I could get going with this system.	☐	☐	☐	☐	☒

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
4. I felt frustrated while using the headset.	☐	☐	☒	☐	☐
5. I found this system to be confusing.	☐	☐	☒	☐	☐
6. Using this system was taxing.	☐	☐	☐	☐	☒
7. The virtual environment is attractive	☒	☐	☐	☐	☐
8. The virtual environment is aesthetically appealing.	☒	☐	☐	☐	☐
9. The virtual environment appealed to my senses.	☒	☐	☐	☐	☐
10. Using this system was worthwhile.	☒	☐	☐	☐	☐

MR

Participant ID _____

021

- | | |
|---|--|
| 1. Which option describes your current home? | ☒ It is owned or being bought by you or someone in household |
| 2. How do you pay for healthcare? | ☒ Private insurance |
| 3. Which best corresponds to your current work situation? | ☒ Not working or looking for work. |

iGROUP Presence Questionnaire MR

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|--|--|
| 1. In the computer-generated world, I had a sense of being there. | ☒ Completely Agree |
| 2. I felt that the virtual world surrounded me. | ☒ Neither Agree nor Disagree |
| 3. I felt like I was perceiving pictures. | ☒ Completely Agree |
| 4. I did not feel present in the virtual space. | ☒ Somewhat disagree |
| 5. I had a sense of acting in the virtual space, rather than operating something from the outside. | ☒ Completely Agree |
| 6. I felt present in the virtual space. | ☒ Somewhat Agree |
| 7. How aware were you of the real world surround you while navigating in the virtual world? | ☒ Very aware |
| 8. I was not aware of my real environment. | ☒ Completely disagree |
| 9. I still paid attention to my real environment. | ☒ Completely Agree |
| 10. I was completely captivated by the virtual world. | ☒ Completely Agree |
| 11. How real did the virtual world seem to you? | ☒ Very real |
| 12. How much your experience in the virtual environment seem consistent with your real-world experience? | ☒ Somewhat Agree |
| 13. How real did the virtual world seem to you? | ☒ Very real |
| 14. The virtual world seemed more realistic than the real world. | ☒ Somewhat Agree |

System Usability Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I think that I would like to use this system frequently.	☐	☐	☐	☐	☒
2. I found the system unnecessarily complex.	☐	☒	☐	☐	☐
3. I thought the system was easy to use.	☐	☐	☐	☐	☒
4. I think that I would need the support of a technical person to be able to use this system.	☒	☐	☐	☐	☐
5. I found the various functions in this system were well integrated.	☐	☐	☐	☐	☒
6. I thought there was too much inconsistency in this system.	☒	☐	☐	☐	☐
7. I would imagine that most people would learn to use this system very quickly.	☐	☐	☐	☐	☒
8. I found the system very cumbersome to use.	☒	☐	☐	☐	☐
9. I felt very confident using the system.	☐	☐	☐	☐	☒
10. I needed to learn a lot of things before I could get going with this system.	☒	☐	☐	☐	☐

User Engagement Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. I lost myself in this mixed-reality experience.	☐	☐	☐	☐	☒
2. The time I spent in mixed reality just slipped away.	☐	☐	☐	☒	☐
3. I was absorbed in the therapy task.	☐	☐	☐	☐	☒
4. I felt frustrated while using the headset.	☒	☐	☐	☐	☐
5. I found this system to be confusing.	☒	☐	☐	☐	☐
6. Using this system was taxing.	☒	☐	☐	☐	☐
7. The virtual environment is attractive	☐	☐	☐	☐	☒

8. The mixed-reality is aesthetically appealing.	☐	☐	☐	☐	☒
9. The blended environment appealed to my senses.	☐	☐	☐	☐	☒
10. Using this system was worthwhile.	☐	☐	☐	☐	☒
11. The therapy experience was rewarding.	☐	☐	☐	☐	☒
12. I felt interested in the therapy tasks.	☐	☐	☐	☐	☒

iPAL Application and Intervention Scale MR

	Completely Disagree	Somewhat Disagree	Neutral	Somewhat Agree	Completely Agree
1. The videos and breathing exercises were a good fit for the strength of my cravings.	☐	☐	☐	☒	☐
2. The videos and breathing exercises matched my internal or external triggers.	☐	☐	☐	☒	☐
4. The breathing exercises were smooth and easy to do.	☐	☐	☐	☒	☐

Exit Interview

1. Which option describes your current home?	☒ It is owned or being bought by you or someone in household
2. How do you pay for healthcare?	☒ Private insurance
3. Which best corresponds to your current work situation?	☒ Not working or looking for work.
1. You experienced Cognitive Behavioral Therapy and Heart Rate Variance interventions in virtual, augmented, and mixed realities. How would you describe your experiences with each of those headsets?	AR - believes that it was easy to navigate and the most simple. enjoyed using this headset a lot. MR - enjoyed the most and felt like it was also very easy to navigate. VR - disliked the most. participant reports that this was very overstimulating and participant reports having a panic attack during the intervention. could not tolerate for greater than 2 minutes.
2. Do you think these types of technologies have a place in OUD treatment and management? If YES, how? If NO, why not?	yes - this could be helpful for individuals who don't socialize much with others and allow for options for other outlets to help with managing cravings.
3. How would you describe the iPAL application, in terms of a cravings management tool?	believes that the app was helpful and covered all important aspects. states that good interventions came were developed based on the answers to her questions.
4. How would you describe the interventions (CBT, HRV)? Were the interventions appropriate for the intensity level and triggers you reported?	CBT - good videos, but felt as if they were a bit monotone and a little too long. believes that they were appropriate for intensity levels reported. HRV - believes that these were too long, but somewhat helpful. believes that both interventions were appropriate for the intensity level and triggers reported.
5. Do you think the iPAL app, or similar digital interventions delivered via phone should be integrated into OUD treatment plans? If YES, how and in what capacity or role? if NO, why not?	yes - believes that the app was adequate and easy to navigate. believes that it was a good cravings management tool.
6. If iPAL or a similar app was available, would you download the app and incorporate it into your OUD treatment and management plan?	doesn't envision self personally using due to having previously been in a significant amount of outpatient treatment. believes that these questions would get redundant and interventions would get boring after a while and participant also believes that they are currently in a good place in recovery to where they don't feel like these current interventions would be very beneficial.